CSC1202 Group 23

Project Scoping

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1. Problem/Opportunity being addressed:

Students in university often have long gaps between their lectures which they would like to use in a productive way by studying or relaxing in a quiet room. Even though DCU's timetables provide information about scheduled classes, there is scarce information about the free rooms available on campus during lecture gaps. Since there are multiple buildings and several rooms on campus that can be used, it would be inconvenient for students to check each room to find a quiet space for their use.

Another challenge we identified is locating lecturers on campus. Students wish to ask quick questions about lecture content to professors, and while it is possible for students to contact professors by email, sometimes it is more efficient to ask a professor questions in person as soon as possible. However, it can be difficult to locate professors on campus since they are busy with lectures. As a result, students do not know which lecture room to find them in. These problems are worthy of addressing because they affect students' daily life and productivity level. If a student spends less time looking for facilities to work in, they will be more inclined to use those facilities and will spend that time working on their projects and coursework and will have more positive university experience overall.

2. Proposed LLM-based agent:

We propose to develop a DCU Campus Locator, an LLM agent that helps DCU students in three ways: provide easy to access timetables, find available study spaces, locate professors on campus. These functionalities will be supported through an interface. The agent will manage these features as follows:

1. Free room locator: The agent will assess the timetables and provide a response to the user with all the available rooms on campus during the requested time.
2. Lecturer locator: Students will be able to search for lecturers by name and check their scheduled lectures, their location and their office location with their published office hours, helping students identify suitable opportunities for in-person consultation.
3. Timetable retrieval: Users will be asked to choose their course code to allow retrieval of their weekly timetable without having to search it manually in MyTimetable website.

The LLM will interpret the students' requests, determine tools or timetable information needed and present relevant results in a clear format.

By putting all these features together into one system, our agent aims to reduce time spent searching for information and making daily navigation activities more convenient for DCU students.

3. Main project goals:

Our main goals for this project are:

1. Improve student experience: Students often feel overwhelmed by the amount of new information. Our project aims to reduce stress by providing them centralised and easy to access platform that offers then places to relax and study, prioritising location, convenience and proximity.
2. Improve lecturer accessibility: Allow students to find lecturers scheduled teaching locations and office hours to help them arrange quick in person consultations
3. Streamline timetable access: Help students quickly access and view weekly lecture timetable by only needing their course code.
4. Centralised location information: Bring timetable, room availability and lecturer scheduling information together in one easy to use platform.
5. Provide personalised experience: Students will be able to interact with the university timetable information and query specific available rooms based on their location and time preferences.